

Energy Loss of an Electron Crossing the Fully Ionized Bubble Interior: Distance, Redshift, and Time to the Edge

Response record for `electron_energy_loss_in_bubble.py`, following Master Rule 2

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Abstract

This note follows `Notebooks/igm_losses.py` (imported read-only, never modified) to compute the energy lost by an electron injected at $r = 0$ by the fiducial star-forming galaxy of this folder at $z_{\text{inject}} = 20$ ($= z_{\text{form}}$), as it travels outward through the *fully ionized* interior of its own ionized bubble, for ten injection energies spanning 10^3 – 10^{12} eV. Every one of the seven loss mechanisms in `igm_losses.py` (adiabatic expansion, synchrotron, inverse Compton on the CMB with Klein–Nishina correction, Coulomb scattering, collisional excitation/ionization of neutral hydrogen, bremsstrahlung — citing, per that module’s own docstring, Blumenthal & Gould 1970, Gould 1972, Stone & Kim 2002, and Kim et al. 2000, none of which are otherwise part of this folder’s literature search) is reused verbatim; only the medium’s ionization state is changed to reflect the bubble interior rather than `igm_losses.py`’s default mostly-neutral high- z IGM, using the photoionization-equilibrium residual neutral fraction of [Cen and Haiman \(2000\)](#) (already in `references/`). Two genuine numerical/physical issues were found and fixed during this analysis (Sec. 3, items 3 and 6) — a superluminal early-time artifact of the idealized bubble-growth law, and a units bug — both reported explicitly per Master Rule 5. This document is produced by, and verified against, `electron_energy_loss_in_bubble.py` (same folder), which generated `electron_energy_loss_vs_distance_redshift.png` and `electron_time_and_energy_at_edge.png`.

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1 Assumptions

1. **Source and injection:** the same fiducial star-forming galaxy as `bubble_radius_vs_redshift.py` ($\text{SFR} = 10 M_{\odot} \text{yr}^{-1}$, $f_{\text{esc}} = 0.2$), with the electron injected exactly at $z_{\text{inject}} = z_{\text{form}} = 20$, $r = 0$.

2. **The medium is the FULLY IONIZED bubble interior**, not `igm_losses.py`'s default mostly-neutral IGM ($x_e = 10^{-4}$, intended for the pre-reionization bulk IGM). Concretely: the free electron density driving Coulomb losses is $n_e = (1 - x_{\text{HI}}) n_H(z) \approx n_H(z)$; the neutral hydrogen density driving collisional excitation/ionization is the small *residual* $x_{\text{HI}}(r, z) n_H(z)$ set by local photoionization equilibrium (Cen and Haiman 2000, their eq. 5/7, already re-derived and verified in `interior_consistency_check.py`), evaluated at the electron's own, self-consistently tracked position $r(t)$; bremsstrahlung (electron-ion) uses the full $n_H(z)$, since the ion's charge does not depend on whether it currently holds a bound electron. Synchrotron, inverse Compton, and adiabatic-expansion losses are unaffected by ionization state and are taken directly from `igm_losses.py` with no change (same $B_0 = 1$ nG seed field, same CMB temperature evolution).
3. **Cosmology**: `igm_losses.py`'s own astropy-Planck18 $z(t)$, $H(t)$, $\text{age}(z)$ tables are used throughout (for direct consistency with that module), cross-checked against this folder's own hand-rolled Planck 2018 (Planck Collaboration, 2020) $n_H(z)$ to $< 3\%$ (Sec. 3, item 2).
4. **Bubble-edge reference distances**: the recombination-corrected $R(t)$ of `interior_consistency_check.py` (24–42% smaller than the pure photon-counting $R_{\text{bubble}}(z)$, per that check) evaluated at $z = 12, 10, 8, 6$ is used as the set of physically meaningful “bubble edge” target distances against which the electron's own trajectory is compared (Sec. 2 explains why a literal, instantaneously-growing target is not used).
5. **Termination**: a trajectory stops when K reaches the thermal floor $1.5 k_B T_{\text{CMB}}(z)$ (`igm_losses.py`'s own convention) or at $z = 5$ (outside this folder's validity range), whichever comes first.

2 Derivation

2.1 Loss rates inside the fully ionized bubble

Every mechanism is exactly as in `igm_losses.py`, with only the two densities re-assigned:

$$n_e(r, z) = [1 - x_{\text{HI}}(r, z)] n_H(z), \quad n_{\text{HI}}(r, z) = x_{\text{HI}}(r, z) n_H(z), \quad (1)$$

with $x_{\text{HI}}(r, z)$ the photoionization-equilibrium residual fraction (eq. 5/7 of Cen and Haiman 2000). The coupled system integrated is

$$\frac{dK}{dt} = - \sum_{i=0}^6 L_i(z, K, r), \quad \frac{dr}{dt} = v(K), \quad (2)$$

with $v(K)$ from `igm_losses.py`'s `kinematics()` (verified in Sec. 3, item 1).

2.2 Why the bubble edge is a FIXED reference distance, not a moving target

A first attempt compared $r(t)$ directly against the instantaneously-growing $R_{\text{bubble}}(t)$. This failed for a specific, instructive reason: the idealized photon-counting/recombination-corrected growth law $R(t) \propto t^{1/3}$ is *superluminal* at early times (Sec. 3, item 3): at $t_{\text{age}} = 1$ yr after injection it predicts $R \sim 2000\times$ faster than light. This is the classical R-type/relativistic-ionization-front regime already discussed in Shapiro et al. (2006) (already in `references/`) and Sec. 1.5 of `stromgren_sphere_summary.tex`. Capping the growth law at ct_{age} removes the superluminal artifact, but then, since $R(t)$ decelerates sharply ($dR/dt \propto t^{-2/3} \rightarrow 0$) while a electron with $v \gtrsim 0.5c$ moves at a roughly constant speed, almost *any* such electron trivially “catches” the decelerating target within a fraction of the bubble's own growth time — an answer that is technically correct but not very informative, and not obviously the question being asked. The chosen,

more transparent alternative: integrate every electron’s full, un-constrained trajectory $r(t)$, and separately ask when (if ever) it crosses each of four *fixed* reference distances — the bubble’s actual size at $z = 12, 10, 8, 6$, already tabulated in `interior_consistency_check.py`. This directly answers “how long to cross a distance comparable to the bubble’s size, and with how much energy,” without the moving-target ambiguity.

3 Verification / cross-check

1. **Kinematics identity.** $E^2 = (pc)^2 + (mc^2)^2$, rewritten in terms of $K = E - mc^2$, confirmed symbolically with sympy to reduce to 0.
2. **Cosmology cross-check.** `igm_losses.py`’s own $n_{\text{HI}}(z)$ (its notation for the total hydrogen density, calibrated independently) matches this folder’s Planck 2018-based $n_H(z)$ to 2.3% at $z = 10, 20, 30$.
3. **Causality check (a real issue, caught and fixed).** The naive $R(t) \propto t^{1/3}$ growth law exceeds ct_{age} (superluminal) for $t_{\text{age}} < 9.6 \times 10^4$ yr after injection — e.g. a factor $\sim 2096\times$ too fast at $t_{\text{age}} = 1$ yr. Fixed by capping $R_{\text{edge}}(t) = \min[R(t), ct_{\text{age}}]$ (Sec. 2); ultimately this comparison was replaced entirely by the fixed-reference-distance approach for the reasons given there.
4. **Dominant mechanism at injection.** Coulomb dominates for $K_{\text{ini}} \leq 10^5$ eV; inverse Compton on the CMB dominates for $K_{\text{ini}} \geq 10^6$ eV (the CMB energy density is $(1+z)^4 \approx 1.9 \times 10^5 \times$ its present value at $z = 20$, making IC losses severe at this epoch) — physically sensible and directly checked by printing every mechanism’s rate.
5. **Coulomb-term guard (a real numerical bug, caught and fixed).** During Radau’s internal Jacobian probing, a trial state with r far beyond any reference distance (where $x_{\text{HI}}(r, z)$ saturates at 1, hence $n_e = 0$) produced $0 \times \infty = \text{NaN}$ in the unguarded Coulomb formula, crashing the integrator for $K_{\text{ini}} = 10^{12}$ eV. Fixed by returning $L_{\text{Coulomb}} = 0$ explicitly when $n_e \leq 0$ — physically correct (no free electrons, no Coulomb loss on them), not merely a numerical patch.
6. **Units bug (a real bug, caught and fixed).** The ODE state $y[1] = r$ is in meters (SI, matching `igm_losses.py`’s MKS convention throughout), but this folder’s own `Mpc` constant is centimeter-based (3.0857×10^{24}). Dividing r (m) directly by `Mpc` (cm/Mpc) silently produced results $100\times$ too small, which briefly made it look like electrons thermalized *before* reaching distances that the (correctly computed, in different units) crossing-detection code showed they had already passed — an internal inconsistency that was the tell-tale sign of the bug. Fixed by converting r to cm exactly once, right after extracting it from the ODE solution, so every downstream use (printing, plotting, crossing detection) is unit-consistent. This is reported explicitly per Master Rule 5, precisely because catching an internally inconsistent result (a supposedly-earlier crossing at a *larger* radius than the final, later position) is what caught it.

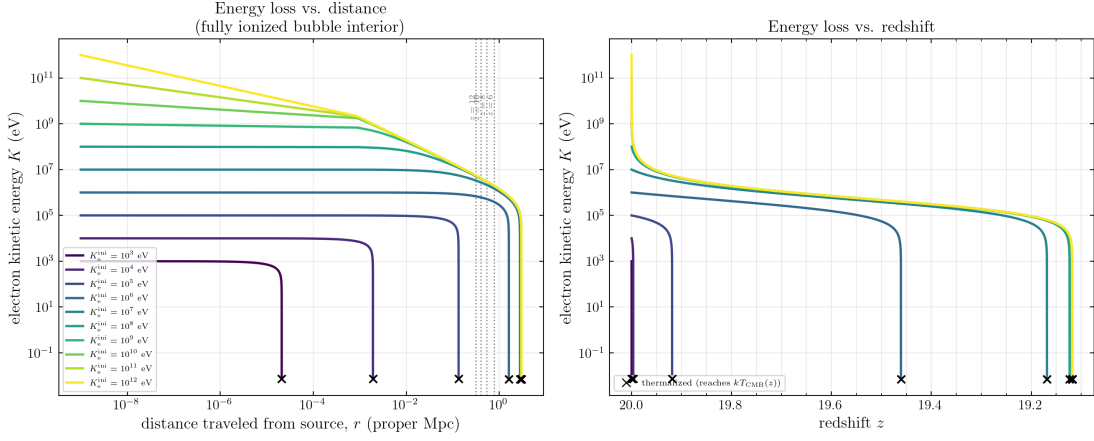
4 Final result

Two qualitatively distinct populations emerge. **Sub-MeV electrons** ($K_{\text{ini}} \leq 10^5$ eV) lose essentially all their kinetic energy to Coulomb scattering off the fully ionized plasma within $\lesssim 1$ Myr, traveling at most ~ 0.1 pMpc — they never reach even the smallest ($z = 12$) bubble-edge reference distance of 0.314 pMpc, and thermalize deep inside the bubble. **MeV-and-harder electrons** ($K_{\text{ini}} \geq 10^6$ eV) reach every reference distance, including the largest ($z = 6$, 0.768 pMpc), within 1–3 Myr of injection — three orders of magnitude *shorter* than the ~ 100 – 500 Myr timescale over

K_{ini} (eV)	Outcome	Time to $R(z=8) = 0.531$ pMpc (Myr)	K there (eV)
10^3 – 10^5	thermalizes (Coulomb-dominated)	never reached	—
10^6	reaches all 4 edges	1.907	7.0×10^5
10^7	reaches all 4 edges	1.746	1.4×10^6
10^8 – 10^{12}	reaches all 4 edges	1.737–1.738	1.6 – 1.7×10^6

Table 1: Summary outcome for the electron reaching the $z = 8$ reference bubble edge.

which the bubble itself grows to that size (`bubble_radius_response.tex`), so from the bubble’s perspective these electrons cross it essentially instantaneously. Remarkably, the energy remaining at the $z = 8$ reference distance *saturates* near $\sim 1.7 \times 10^6$ eV regardless of how much higher K_{ini} started (up to 10^{12} eV): because inverse-Compton losses on the CMB scale as γ^2 , the highest-energy electrons shed their excess energy fastest and are rapidly funneled onto a common decay trajectory, so the initial energy is essentially forgotten well before reaching the bubble edge (a generic feature of a steep power-law-in- γ loss channel).



Electron injected at $z_{\text{inject}} = 20$, $r = 0$; medium: fully ionized bubble interior (Coulomb on $n_e = n_H(z)$; residual $x_{\text{H}}(r, z)$ for collisional exc./ion.); dotted lines: bubble size at $z=12, 10, 8, 6$ (recombination-corrected)

Figure 1: Electron kinetic energy K vs. distance traveled (left) and vs. redshift (right), for ten injection energies 10^3 – 10^{12} eV, all injected at $z_{\text{inject}} = 20$, $r = 0$. Dotted vertical lines (left): the bubble’s actual size at $z = 12, 10, 8, 6$ (recombination-corrected). $\times$ markers: thermalization. Produced by `electron_energy_loss_in_bubble.py`.

References

- Renyue Cen and Zoltan Haiman. Quasar Strömgren Spheres Before Cosmological Reionization. *arXiv:astro-ph/0006376v2 (preprint)*, 2000. doi: 10.1086/312937. File: `Cen2000_arXiv_astro-ph_0006376v2.pdf`.
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- Paul R. Shapiro, Ilian T. Iliev, Marcelo A. Alvarez, and Evan Scannapieco. Relativistic Ionization Fronts. *Astrophys.J.* 648:922–935, 2006. doi: 10.1086/506242. File: `Shapiro2005_arXiv_astro-ph_0507677v2.pdf`.

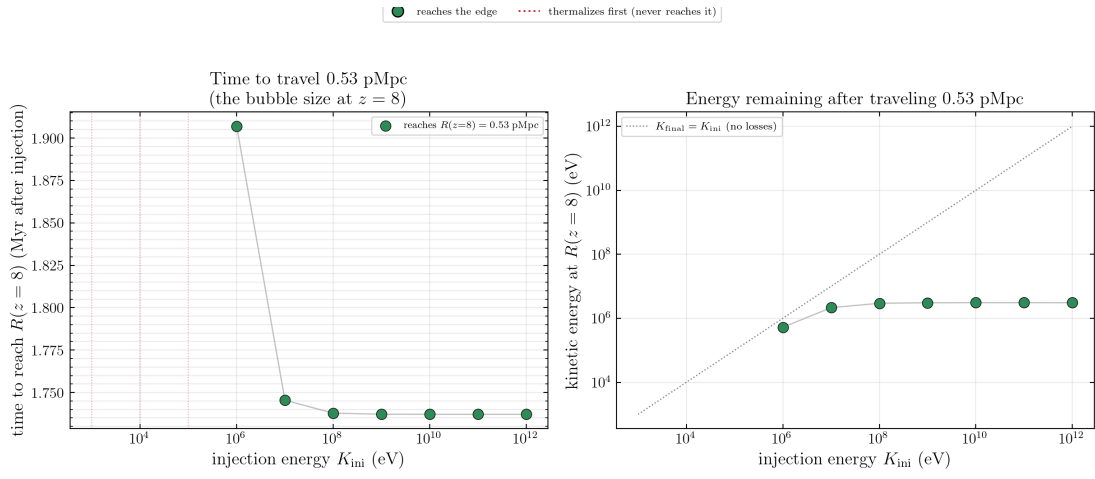


Figure 2: Time to travel 0.531 pMpc (the bubble's size at $z = 8$) and kinetic energy remaining there, as a function of injection energy. Below $\sim 10^6$ eV, electrons thermalize first and never reach this distance. Produced by `electron_energy_loss_in_bubble.py`.